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🔍 Type ↵ to search

8

+

<> Code

🔍 Issues

🔗 Pull requests

👤 Agents

🎬 Actions

📁 Projects

📖 Wiki

🛡 Security and quality

📈 Insights

⚙ Settings

coit20246y26t1-project-coit20246-sydb1-masoodjazib-1 / plan.md

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Masood3081

Update plan.md

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Preview

Code

Blame

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Project Plan

Project Details

Item	Details
Unit	COIT20246 Cyber Security and Networking
Project	OpenWRT Small Business Network Security Project
Group Number	13
Campus	SYD
Tutor	Ahmad Saeed
Student 1	Jazib Hussain Syed

Item	Details
Student 1 ID	12311998
Student 2	Masood Ali Syed
Student 2 ID	12316372

Group Members

Name	Student ID	Main Responsibility
Jazib Hussain Syed	12311998	OpenWRT setup, network documentation, firewall configuration, screenshots
Masood Ali Syed	12316372	Hardening, traffic analysis, risk assessment, security controls, reflection

Communication Plan

The group will communicate through WhatsApp, Zoom, and tutorial discussions. WhatsApp will be used for quick updates, sharing screenshots, discussing configuration issues, and confirming GitHub updates. Zoom will be used when both students need to review the OpenWRT setup, Wireshark captures, diagrams, report files, or demonstration video together. Tutorial time will be used to clarify technical issues with the tutor where required.

The group will aim to communicate at least twice per week. One discussion will take place during or after tutorial time, and one additional online discussion will be arranged when practical work or report writing needs to be reviewed. Both students will check the GitHub repository regularly to ensure that screenshots, Markdown content, diagrams, packet capture files, and risk assessment files are updated correctly.

Project Schedule

Week	Planned Work
Week 5	Form the group, access the GitHub repository through GitHub Classroom, review the project specification, and prepare the project plan.
Week 6	Document the OpenWRT lab setup, identify OpenWRT interfaces, record IP addresses, check VirtualBox adapter settings, and begin the lab network diagram.
Week 7	Configure the OpenWRT-hosted test website, verify access from the Windows host, and complete HTTP, SSH, ICMP, and management interface firewall demonstrations.
Week 8	Complete OpenWRT hardening tasks, including root password change, password hash review, SSH key-based authentication, and disabling unnecessary services.
Week 9	Capture and analyse HTTP and SSH traffic using tcpdump and Wireshark, complete the TVAMatrix risk assessment, and recommend appropriate security controls.
Week 10	Finalise Markdown report files, check image and file links, complete reflection, export PDFs, prepare the ZIP file, and record the 3–5 minute demonstration video.

Task Allocation

Project Area	Task	Responsible Student	Evidence or Output
Planning	Project plan and communication schedule	Both students	plan.md
Business scenario	Business assumptions and website purpose	Both students	network.md

Project Area	Task	Responsible Student	Evidence or Output
Lab network	OpenWRT interface identification and VirtualBox adapter documentation	Jazib Hussain Syed	Screenshots in <code>images/</code>
Lab network diagram	Draw lab network diagram showing OpenWRT, Windows host, interfaces, adapter types, and IP addresses	Jazib Hussain Syed	<code>.drawio</code> file and exported image
Website setup	Create and test the OpenWRT-hosted business website	Jazib Hussain Syed	Browser screenshot
Firewall rules	Demonstrate HTTP, SSH, ICMP, and management interface rules with before-and-after evidence	Jazib Hussain Syed	Firewall and test screenshots
Production design	Create production network diagram and IP allocation table	Both students	<code>.drawio</code> file, exported image, and IP table
Hardening	Change root password, check <code>/etc/shadow</code> , configure SSH keys, and disable unnecessary service	Masood Ali Syed	Hardening screenshots and explanation
Traffic analysis	Capture and analyse HTTP and SSH traffic using tcpdump and Wireshark	Masood Ali Syed	<code>.pcap</code> files and Wireshark screenshots
Risk assessment	Complete mini cyber security risk assessment using TVAMatrix	Both students	Risk assessment spreadsheet and screenshots
Security controls	Recommend three controls for the highest-risk data asset	Both students	Written explanation in <code>security.md</code>
Reflection	Discuss contribution, commits, task split, teamwork issues, and future improvements	Both students	<code>reflection.md</code> and GitHub commit screenshot

Project Area	Task	Responsible Student	Evidence or Output
Demonstration video	Record firewall rule demonstration and packet capture walkthrough	Both students	Single MP4 submitted through Echo360

GitHub Contribution Plan

Both students will commit their own work to the GitHub repository. Commits will be made after meaningful updates such as writing Markdown sections, uploading screenshots, adding diagrams, uploading packet capture files, updating the risk assessment, or revising the report. One student will not commit work on behalf of the other student.

This contribution plan is used to show individual participation and maintain a clear record of project progress. Each student will aim to make regular commits across the project period rather than leaving all updates until the final submission stage.

File Organisation Plan

File or Folder	Purpose
README.md	Main project overview, report links, and checklist
plan.md	Group plan, communication plan, schedule, and task allocation
network.md	Business assumptions, lab setup, firewall rules, diagrams, and IP allocation
harden.md	OpenWRT hardening tasks, screenshots, and explanations
security.md	Traffic analysis, risk assessment, and recommended security controls
reflection.md	Group contribution, GitHub commits, teamwork reflection, and lessons learned
images/	Screenshots and exported diagram images

File or Folder	Purpose
<code>captures/</code>	HTTP and SSH packet capture <code>.pcap</code> files
<code>risk-assessment.xlsx</code>	Completed TVAMatrix risk assessment spreadsheet

Professional Practice

The group will follow the project requirements by using GitHub, keeping Markdown files organised, creating original screenshots from the group’s own virtual environment, preparing original diagrams in draw.io, and maintaining consistent file names. The final Moodle submission will include PDF versions of the Markdown files, a ZIP of the full GitHub repository, and a demonstration video submitted through Echo360.

The group will also ensure that the GitHub repository, Moodle ZIP file, and PDF files contain the same final version of the project work before submission.